INTRODUCTION TO MALE REPRODUCTIVE SYSTEM

MALE REPRODUCTIVE SYSTEM

- Consists of a number of sex organs that are a part of the human reproductive process.

- Produces, stores and releases the mature gametes or sperms.

MALE REPRODUCTIVE ORGANS

External genital organs:

- Penis
- Scrotum

Internal genital organs:

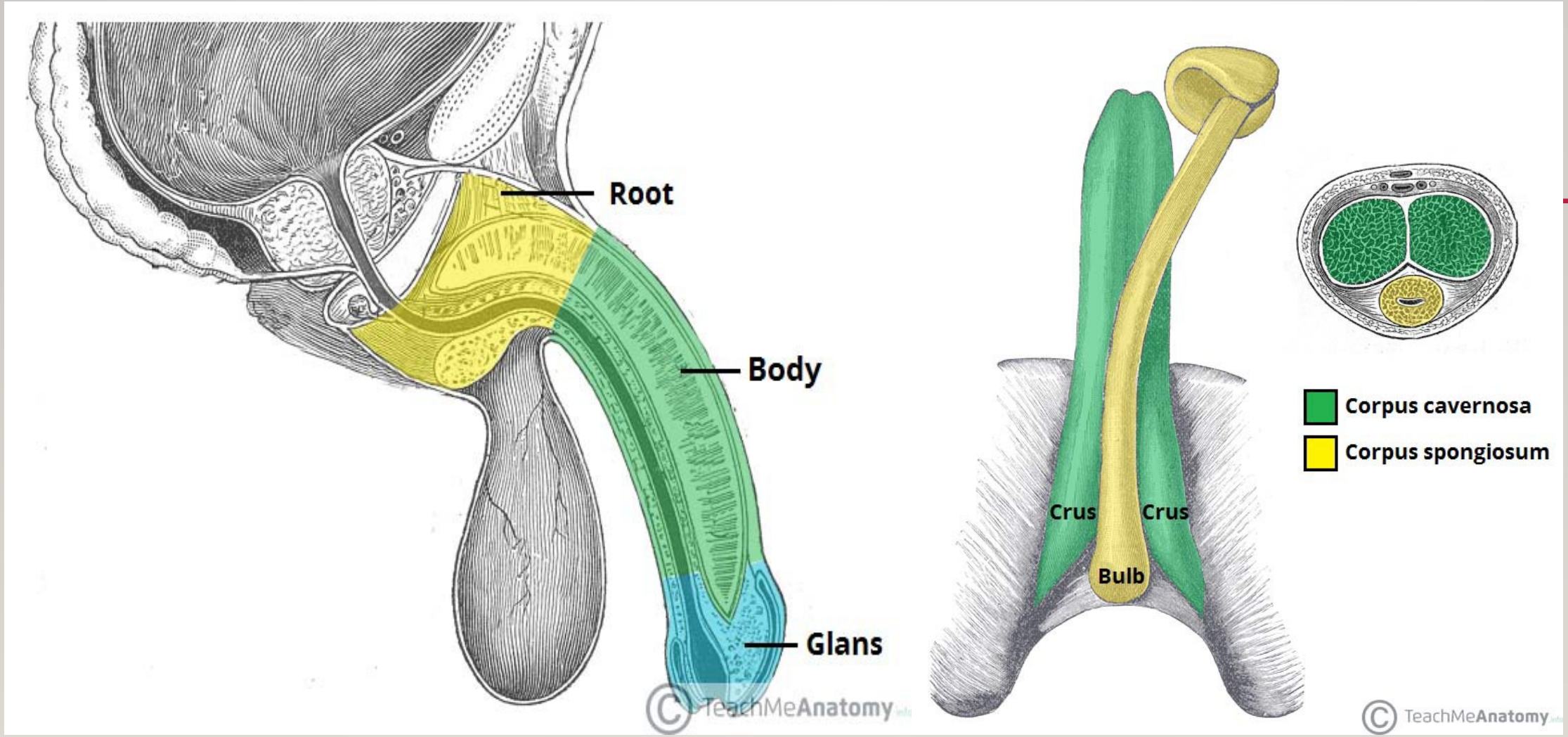
- Testis
- Epididymis
- Vas Deferens
- Prostate
- Seminal vesicles and bulbourethral glands

THE PENIS

- The penis is an external organ of the male reproductive system.

STRUCTURE:

- The penis can be anatomically divided into three parts:
- **Root** – the most proximal, fixed part of the penis. It is located in the superficial perineal pouch of the pelvic floor, and is not visible externally. The root contains three erectile tissues (two crura and bulb of the penis), and two muscles (ischiocavernosus and bulbospongiosus)
- **Body**: the free part of the penis, located between the root and glans. It is suspended from the pubic symphysis. It is composed of three cylinders of erectile tissue – two corpora cavernosa, and the corpus spongiosum
- **Glans:** the most distal part of the of penis. It is conical in shape, and is formed by the distal expansion of the corpus spongiosum. This contains the opening of the urethra, termed the external urethral orifice.


Root
Body
Glans
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Crus
Crus
Bulb
Corpus cavernosa
Corpus spongiosum
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MUSCLES:

- There are four muscles located in the root of the penis:
- **Bulbospongiosus** (x2) – associated with the bulb of the penis. It contracts to empty the spongy urethra of any residual semen and urine. The anterior fibres also aid in maintaining erection by increasing the pressure in the bulb of the penis.
- **Ischiocavernosus** (x2) – surrounds the left and right crura of the penis. It contracts to force blood from the cavernous spaces in the crura into the corpus cavernosa – this helps maintain erection.

FASCIAL COVERINGS:

- Each mass of erectile tissue has two fascial coverings.
- The most superficial layer, immediately under the skin, is the external fascia of **Colles**
- A deeper stratum is the **deep fascia of the penis** (also known as **Buck's fascia**).
- Underneath the deep fascia is the strong fascia called **tunica albuginea**, forming an individual capsule

Ligaments:

- The root of the penis is supported by two ligaments, which attach it to the surrounding structures:
- **Suspensory ligament** – a condensation of deep fascia. It connects the erectile bodies of the penis to the pubic symphysis.
- **Fundiform ligament** – a condensation of abdominal subcutaneous tissue. It runs down from the linea alba, surrounding the penis like a sling, and attaching to the pubic symphysis.

VASCULATURE

The penis receives arterial supply from three sources:

- Dorsal arteries of the penis
- Deep arteries of the penis
- Bulbourethral artery

These arteries are all branches of the **internal pudendal artery**.

The venous drainage is through:

- **superficial dorsal veins**
- **Deep dorsal vein of penis**

NERVE SUPPLY

- The penis is supplied by **S2-S4** spinal cord segments and spinal ganglia
- sympathetic innervation by dorsal nerve of the penis, a branch of the **pudendal nerve**.
- Parasympathetic innervation is carried by cavernous nerves from the **peri-prostatic nerve plexus.**
- **Erection** is by parasympathetic and **Ejaculation** is by sympathetic system.

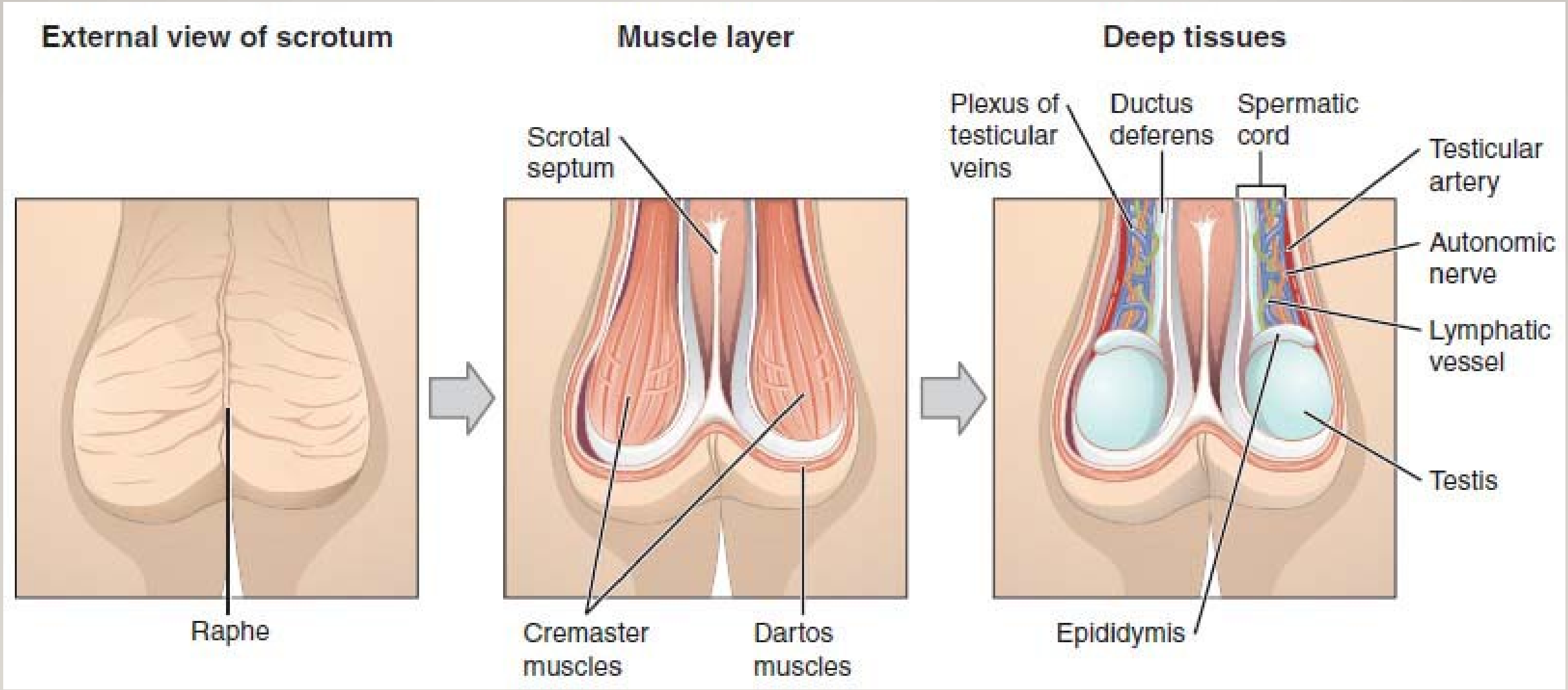
SCROTUM

The scrotum is a **fibromuscular** cutaneous sac, located between the penis and anus. It is dual-chambered, forming an expansion of the perineum.

The scrotum contains three major (paired) structures:

- **Testis** – the site of sperm production.
- **Epididymis** – situated at the head of each testicle. It functions as a storage reservoir for sperm.
- **Spermatic cord** – a collection of muscle fibres, vessels, nerves and ducts that run to and from the testes.

- There are also muscle fibres located within the scrotum. The **dartos** muscle is a sheet of smooth muscle, situated immediately underneath the skin. It acts to help regulate the temperature of the scrotum, by **wrinkling** the skin – this decreases surface area, reducing heat loss.

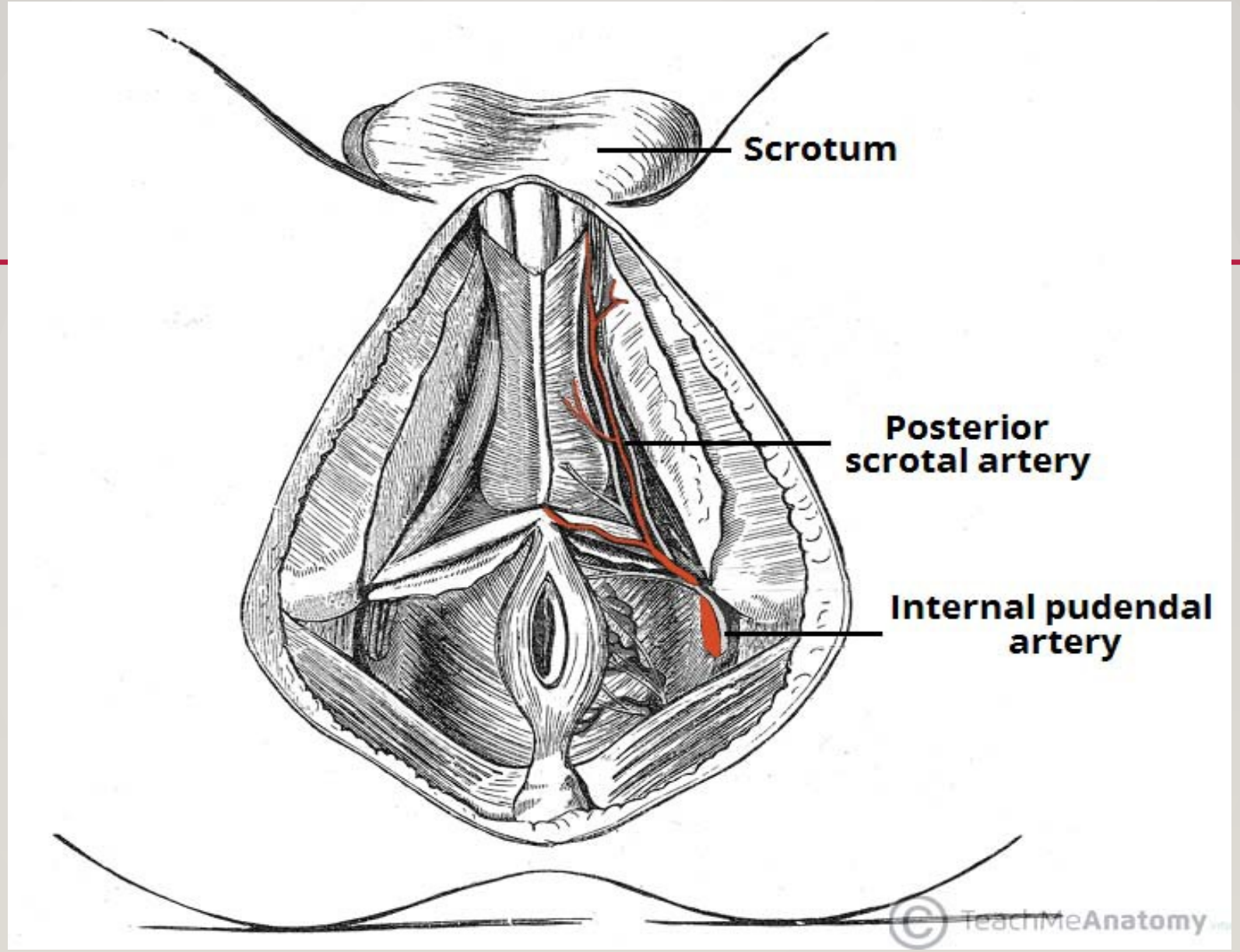

External view of scrotum
Muscle layer
Deep tissues
Raphe
Scrotal septum
Cremaster muscles
Dartos muscles
Plexus of testicular veins
Ductus deferens
Spermatic cord
Testicular artery
Autonomic nerve
Lymphatic vessel
Testis
Epididymis

VASCULATURE

The scrotum receives arterial supply from:

- **Anterior scrotal artery** (branch of external pudendal artery)
- **posterior scrotal artery** (branch of internal pudendal artery)

- Nerve supply: The **scrotal veins** follow the major arteries, draining into the external pudendal veins.


Scrotum
Posterior scrotal artery
Internal pudendal artery
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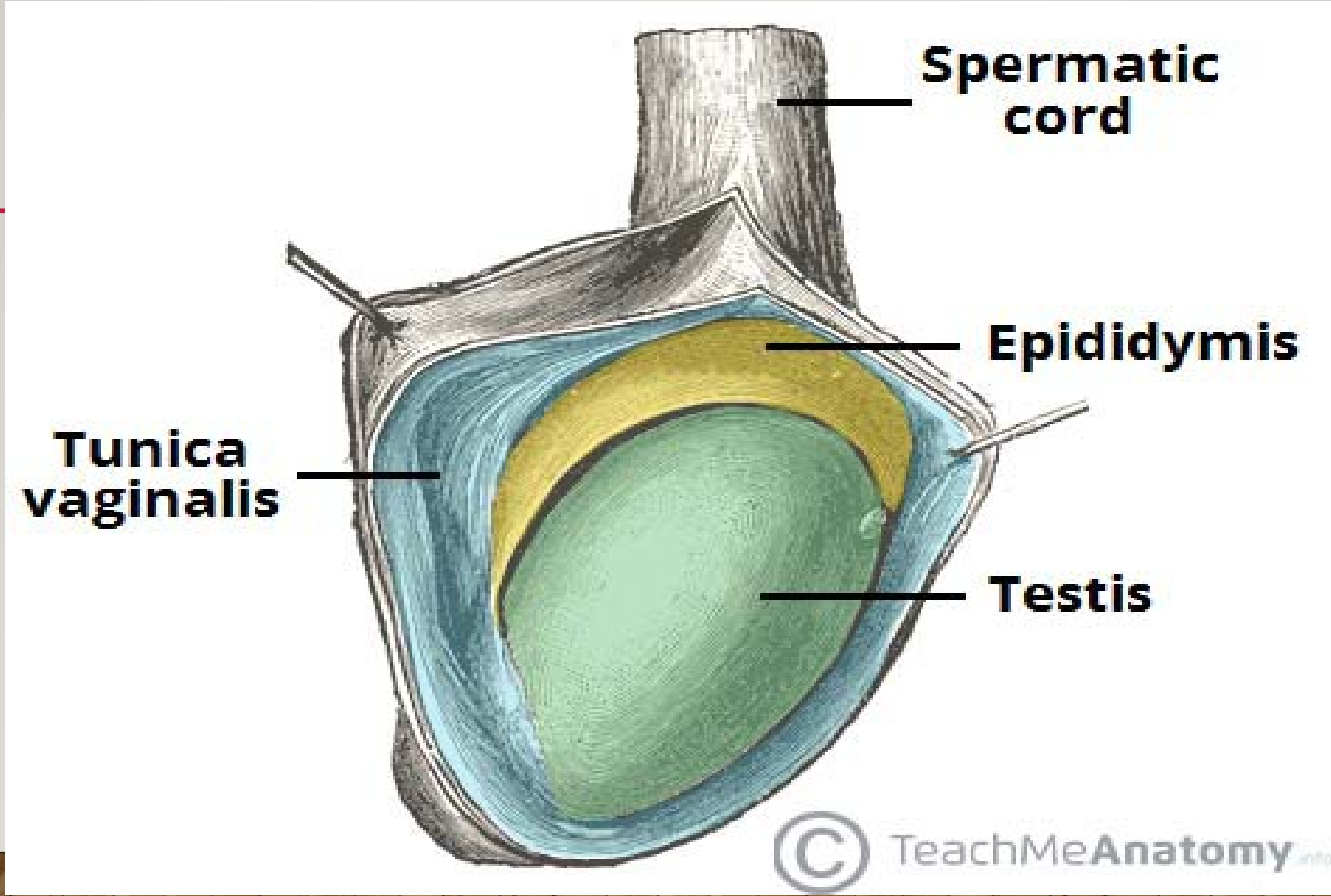
NERVE SUPPLY:

- **Anterior and anterolateral aspect** – Anterior scrotal nerves derived from the genital branch of genitofemoral nerve and ilioinguinal nerve**.**
- **Posterior aspect** – Posterior scrotal nerves derived from the perineal branches of the internal pudendal nerve and posterior femoral cutaneous nerve.

LYMPHATICS: The lymphatic fluid from the scrotum drains to the nearby **superficial inguinal nodes**.

TESTES, EPIDIDYMIS, DUCTUS DEFERENCE, SEMINAL VESICLES

- The **testes and epididymis** are paired structures, located within the scrotum. The testes are the site of sperm production and hormone synthesis, while the epididymis has a role in the storage of sperm.
- The testes are located within the **scrotum**, with the epididymis situated on the posterolateral aspect of each testicle. Commonly, the left testicle lies lower than the right. They are suspended from the abdomen by the **spermatic cord**.
- Originally, the testes are located on the **posterior abdominal wall**. During embryonic development they descend down the abdomen, and through the inguinal canal to reach the scrotum. They carry their neurovascular and lymphatic supply with them.


Spermatic cord
Epididymis
Tunica vaginalis
Testis
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ANATOMICAL STRUCTURE

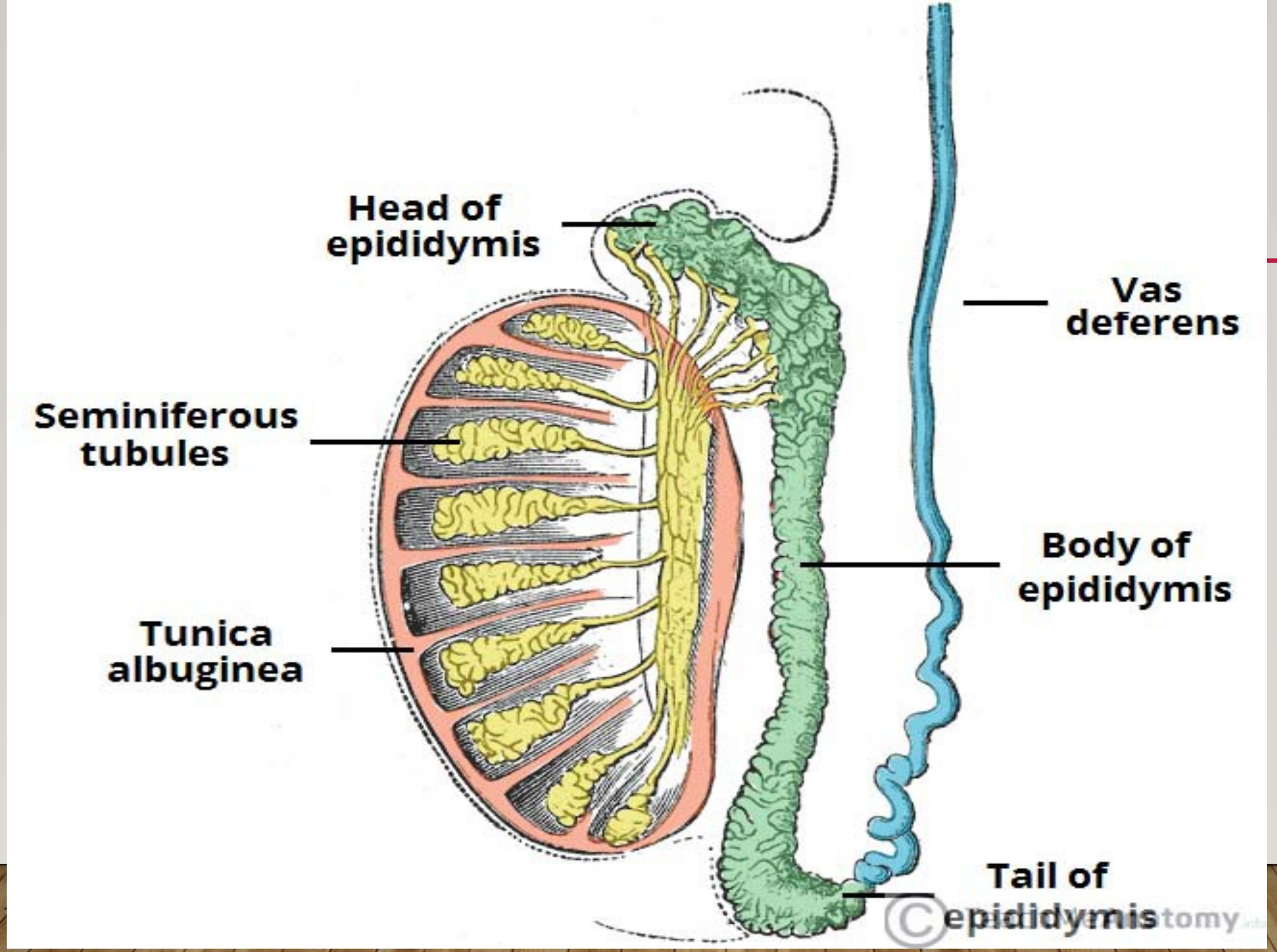
TESTES:

- The testes have an ellipsoid shape. They consist of a series of lobules, each containing **seminiferous tubules** supported by interstitial tissue.
- The seminiferous tubules are lined by Sertoli cells that aid the maturation process of the spermatozoa. In the interstitial tissue lie the Leydig cells that are responsible for testosterone production.
- Inside the scrotum, the testes are covered almost entirely by the **tunica vaginalis**, a closed sac of parietal peritoneal origin that creates a thin sac containing a small amount of viscous fluid. This sac covers the anterior surface and sides of each testicle and works much like the peritoneal sac, lubricating the surfaces of the testes and allowing for friction-free movement.

- The testicular parenchyma is protected by the tunica albuginea, a fibrous capsule that encloses the testes. It penetrates into the parenchyma of each testicle with diaphragms, dividing it into lobules.

EPIDIDYMIS:

- The epididymis consists of a single heavily coiled duct. It can be divided into three parts; head, body and tail.
- Head – The most proximal part of the epididymis. It is formed by the efferent tubules of the testes, which transport sperm from the testes to the epididymis.
- Body – Formed by the heavily coiled duct of the epididymis.
- Tail – The most distal part of the epididymis. It marks the origin of the vas deferens, which transports sperm to the prostatic portion of the urethra for ejaculation.

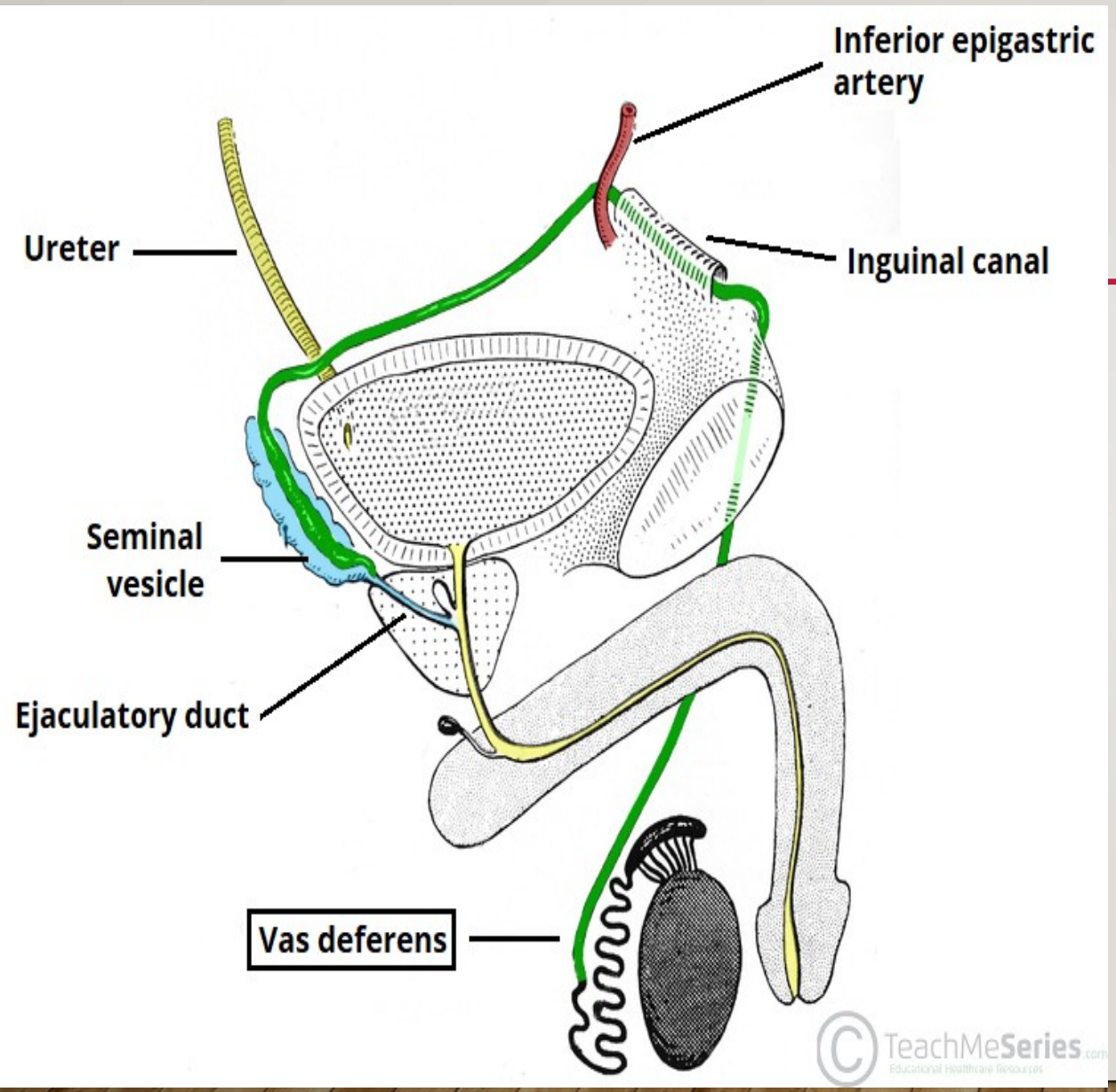

Head of epididymis
Vas deferens
Seminiferous tubules
Body of epididymis
Tunica albuginea
Tail of epididymis

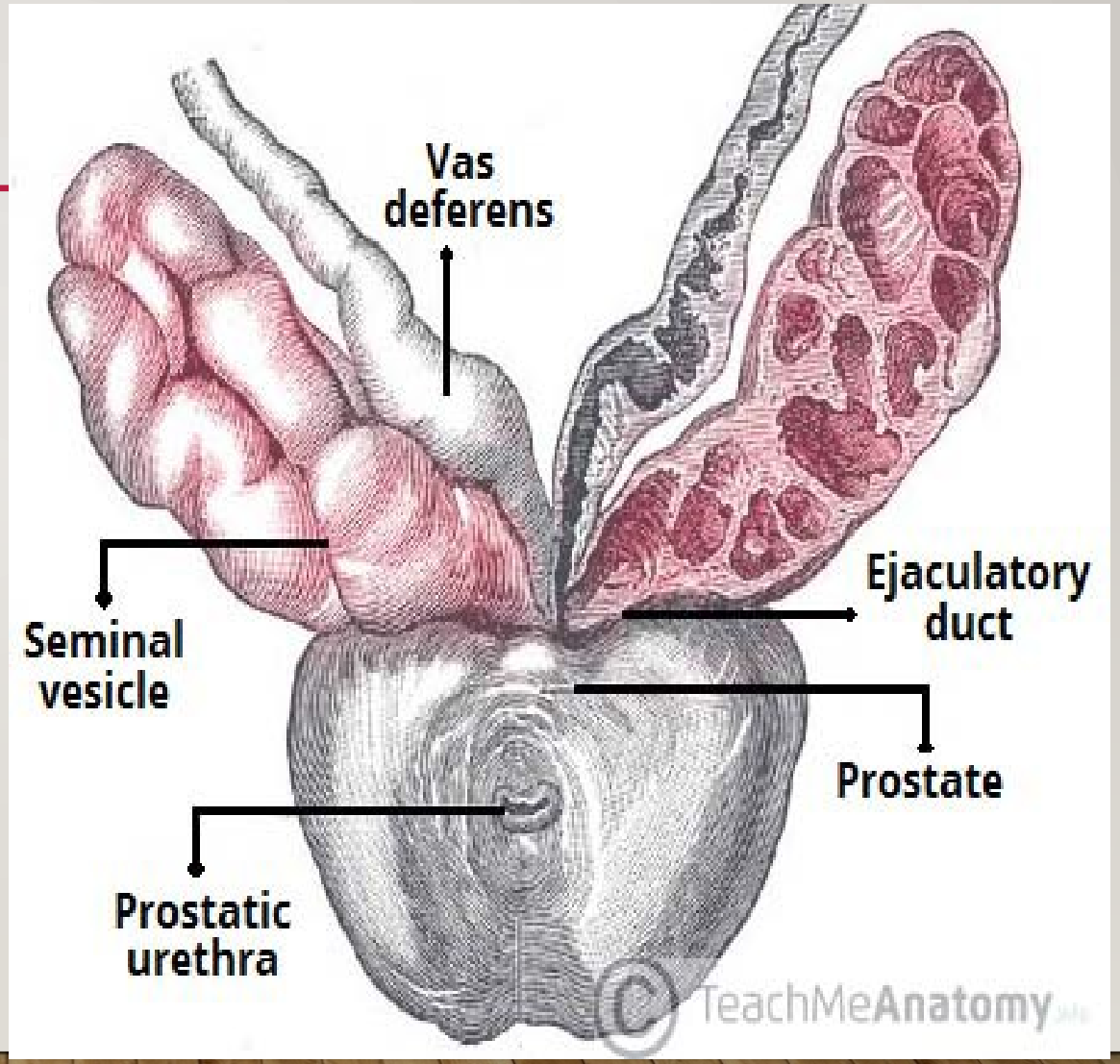
Vas Deferens:

- the vas deferens is a straight, thick muscular tube that conveys sperm from the **epididymis** to the ampulla and eventually, to the ejaculatory duct
- Ejaculatory duct is formed by the convergence of the vas deferens and seminal vesicle duct.
- From the ejaculatory duct, sperm can pass through to the prostatic urethra.

SEMINAL VESICLES:

- The **seminal vesicles** (also known as the vesicular or seminal glands) are a pair of 5cm long tubular glands. They are located between the **bladder fundus** and the **rectum.**
- Their most important anatomical relation is with the vas deferens, which combine with the duct of the seminal vesicles to form the **ejaculatory duct**, which subsequently drains into the prostatic urethra.

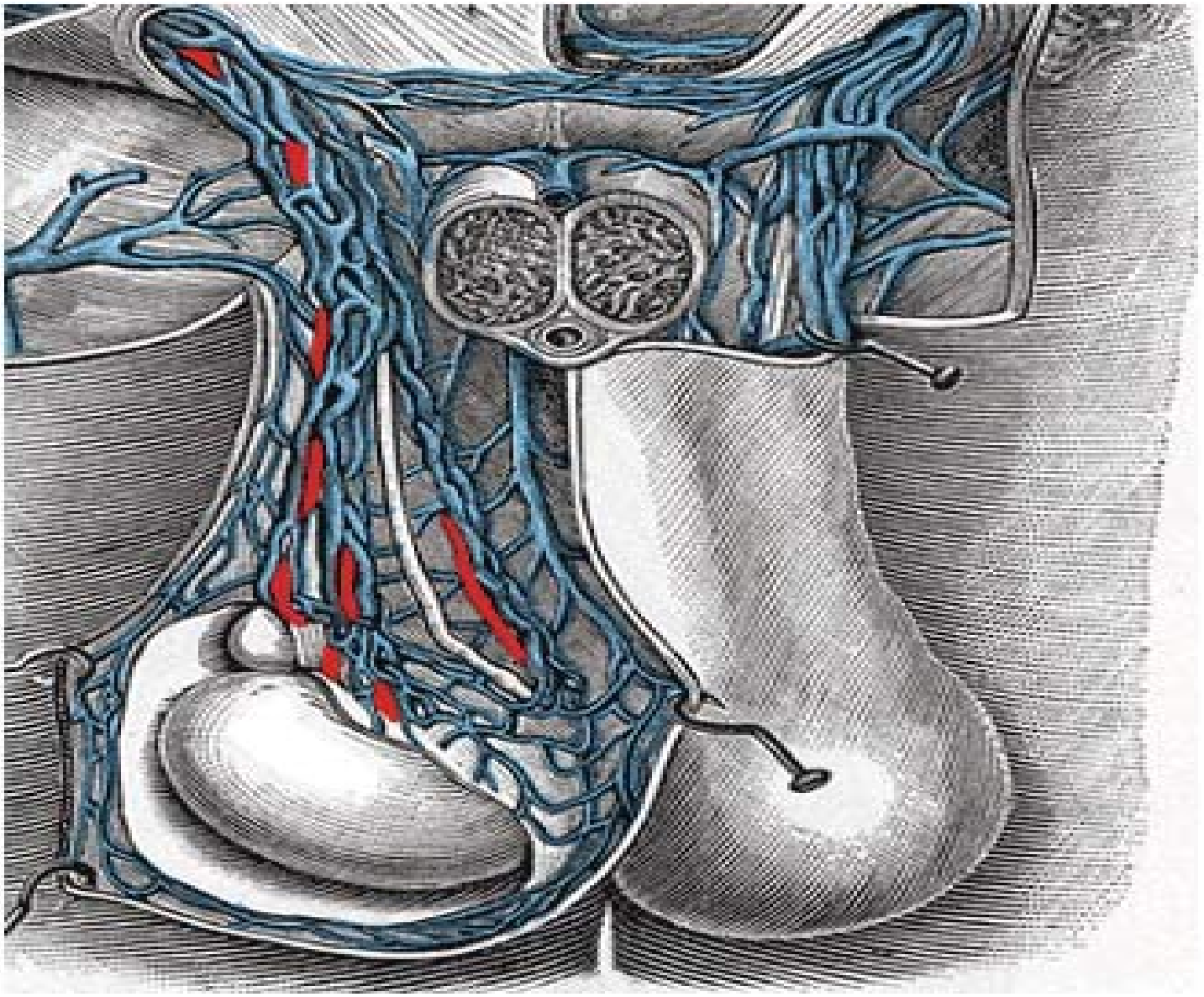

Inferior epigastric artery
Ureter
Inguinal canal
Seminal vesicle
Ejaculatory duct
Vas deferens
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Vas deferens
Seminal vesicle
Ejaculatory duct
Prostate
Prostatic urethra
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VASCULATURE

- The main arterial supply to the testes and epididymis is via the paired **testicular arteries,** which arise directly from the abdominal aorta.

- Venous drainage is achieved via the paired testicular veins. They are formed from the **pampiniform plexus** in the scrotum – a network of veins wrapped around the testicular artery.



NERVE SUPPLY:

- The testes and epididymis receive innervation from the **testicular plexus** – a network of nerves derived from the renal and aortic plexi.

LYMPHATICS:

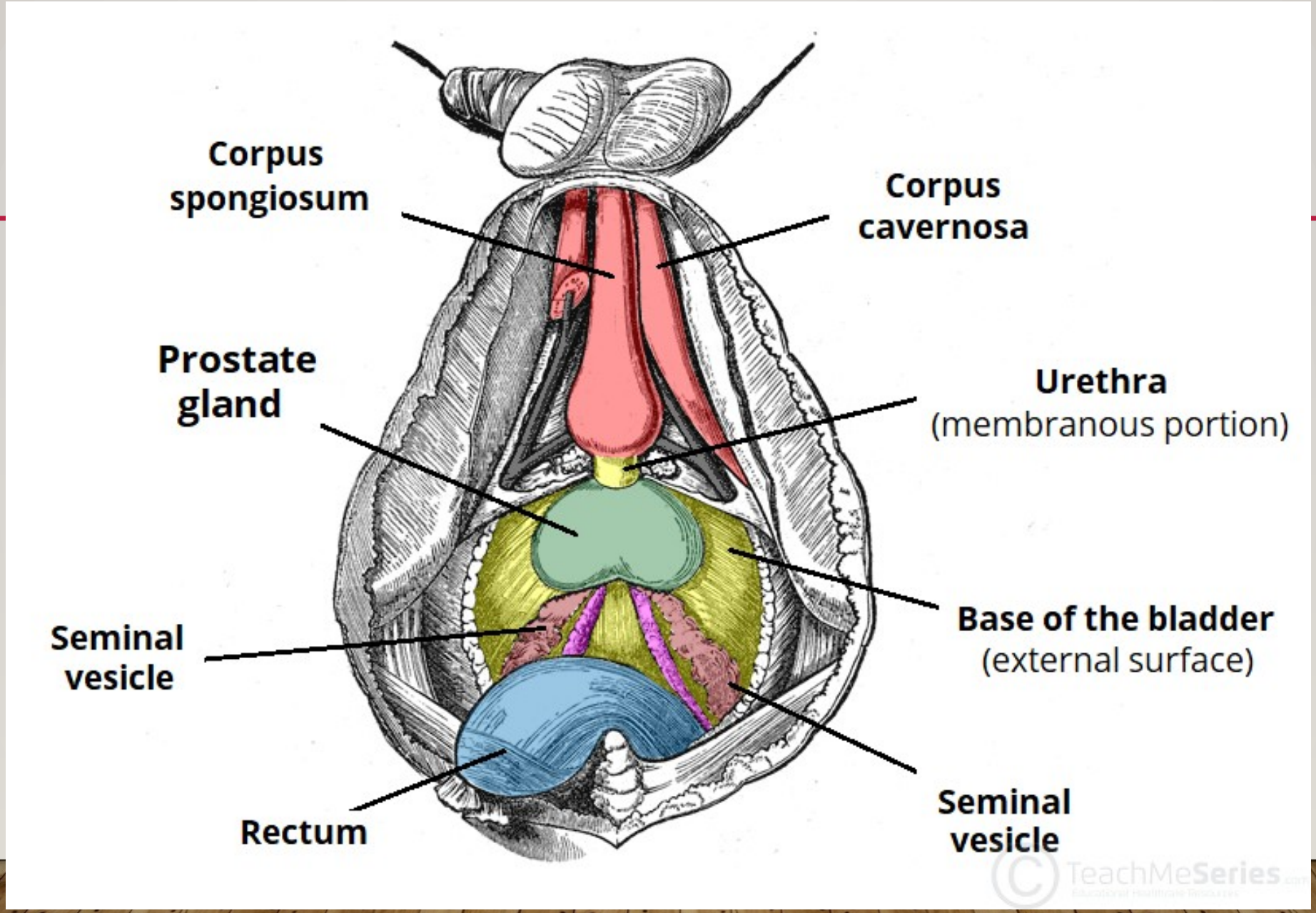
- the lymphatic drainage is to the **lumbar** and **para-aortic nodes.**

THE PROSTATE

- The prostate is the largest accessory gland in the male reproductive system.It secretes proteolytic enzymes into the semen, which act to break down clotting factors in the ejaculate. This allows the semen to remain in a fluid state, moving throughout the female reproductive tract for potential fertilisation.

ANATOMICAL POSITION

- The prostate is positioned inferiorly to the neck of the bladder and superiorly to the **external urethral sphincter**, with the levator ani muscle lying inferolaterally to the gland.
- Most importantly, posteriorly to the prostate lies the ampulla of the **rectum** – this anatomical arrangement is utilised during Digital Rectal Examinations (DRE), allowing physicians to examine the gland.

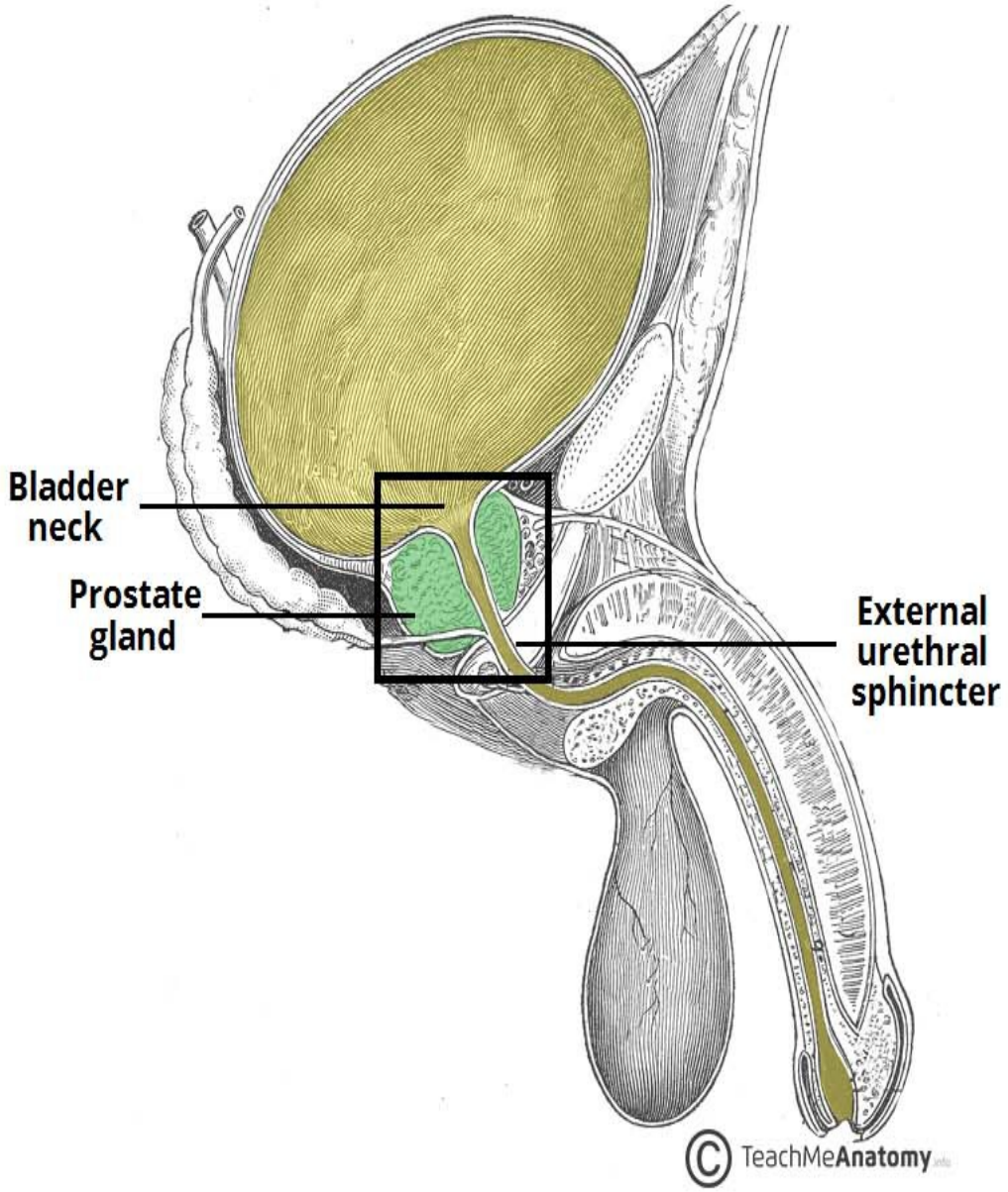

Corpus spongiosum
Corpus cavernosa
Prostate gland
Urethra (membranous portion)
Seminal vesicle
Base of the bladder (external surface)
Rectum
Seminal vesicle
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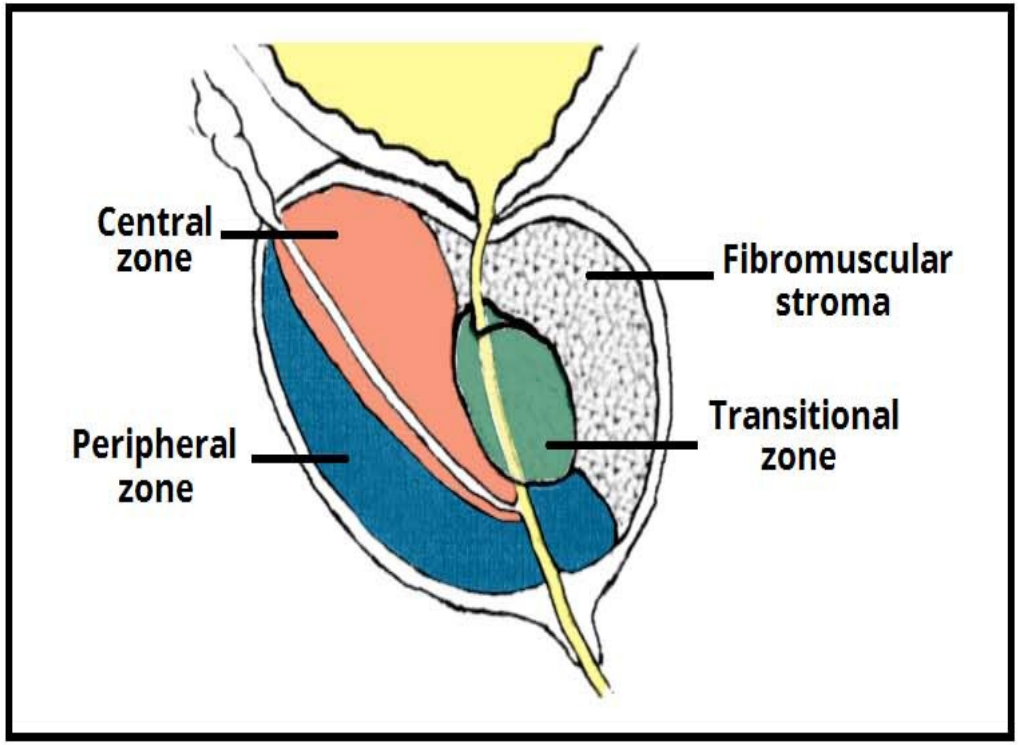
- The prostate is commonly described as being the size of a **walnut**. Roughly two-thirds of the prostate is glandular in structure and the remaining third is fibromuscular. The gland itself is surrounded by a thin fibrous capsule of the prostate.
- the prostate is divided into anatomical lobes (inferoposterior, inferolateral, superomedial, and anteromedial) by the urethra and the ejaculatory ducts as they pass through the organ.

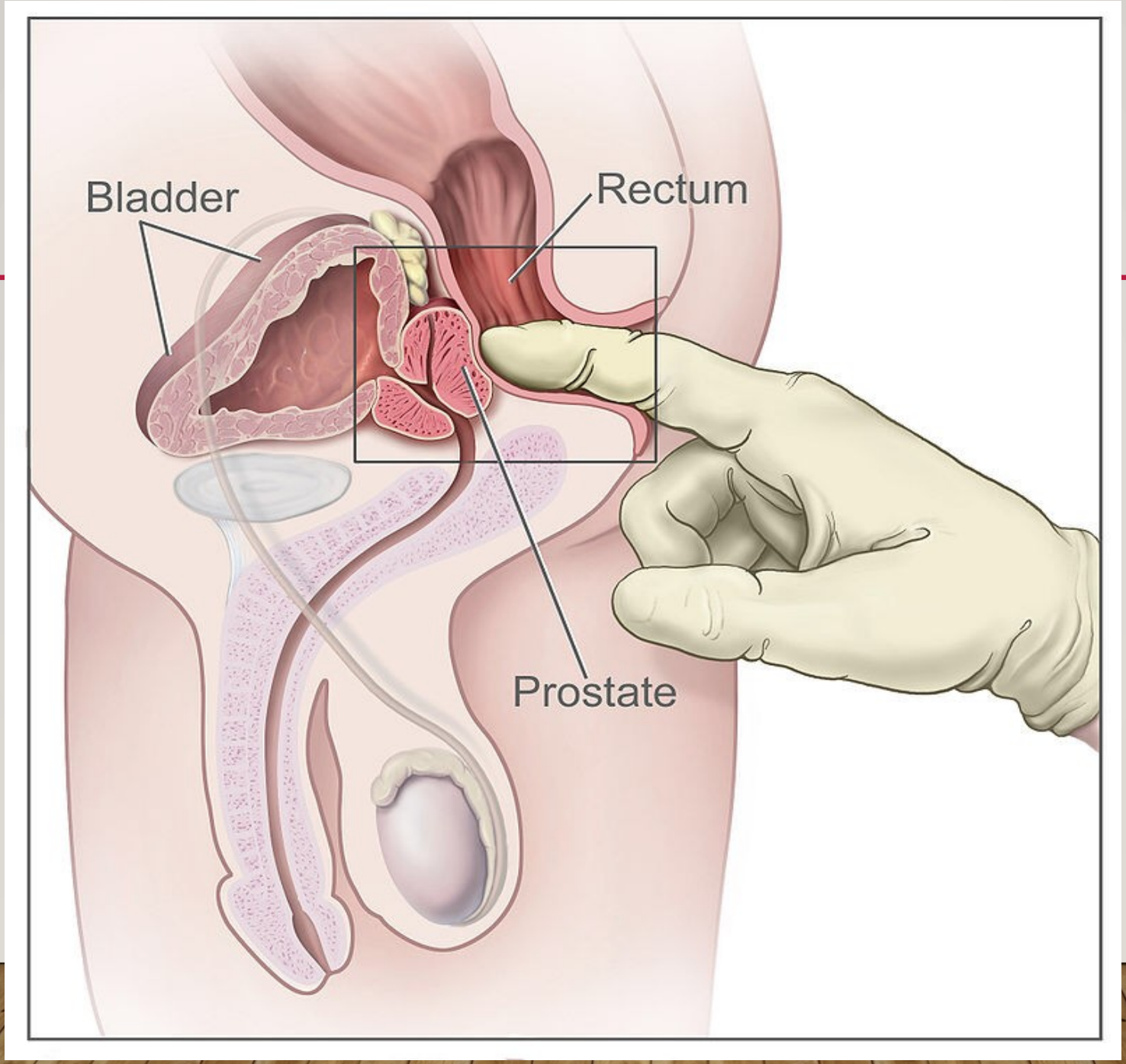
- However, more important clinically is the histological division of the prostate into **three zones:**
  - **Central zone** – surrounds the ejaculatory ducts, comprising approximately 25% of normal prostate volume.
  - **Transitional zone** – located centrally and surrounds the urethra, comprising approximately 5-10% of normal prostate volume. The glands of the transitional zone are those that typically undergo benign hyperplasia (BPH)

- **Peripheral zone** – makes up the main body of the gland (approximately 65%) and is located posteriorly. It is linked to the high incidence of prostate carcinoma at the peripheral zone.The peripheral zone is mainly the area felt against the rectum on DRE, which is of irreplaceable value.

- The **fibromuscular stroma** (or fourth zone for some) is situated anteriorly in the gland. It merges with the tissue of the urogenital diaphragm. This part of the gland is actually the result of interaction of the prostate gland budding around the urethra during prostate embryogenesis and the common horseshoe-like muscle precursor of the smooth and striated muscle that will eventually form the internal and external urethra sphincter.


Bladder neck
Prostate gland
External urethral sphincter
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Central zone
Fibromuscular stroma
Peripheral zone
Transitional zone


Bladder
Rectum
Prostate

VASCULATURE

- The arterial supply to the prostate comes from the **prostatic arteries**, which are mainly derived from the internal iliac arteries.

- Venous drainage of the prostate is via the **prostatic venous plexus**, draining into the internal iliac veins.

- **NERVE SUPPLY:** The prostate receives sympathetic, parasympathetic and sensory innervation from the **inferior hypogastric plexus**.

THANK YOU